Gokhale Education Society's

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Research on Employee Performance Prediction Based on Machine Learning

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Flow of Presentation

1. Introduction
2. Literature Survey
3. Concepts / Theory
   - 3.1 Details of Architecture/ Algorithms (if any)
   - 3.2 Description of Models /Methods
   - 3.3 Advantages and Disadvantages
   - 3.4 Applications/Uses
4. Conclusion
5. References

Introduction

- Employee performance is a key factor for organizational growth and success.
- Traditional performance evaluation methods are often time-consuming and biased.
- Machine Learning (ML) provides data-driven, objective insights.
- ML models can analyze employee data to predict future performance.
- These predictions help HR in making better decisions.
- ML helps streamline HR operations and maintain fairness.
- Manual evaluation is often time-consuming

Literature Survey

- Data was gathered from real or synthetic HR datasets.
- Accuracy ranged from 70% to 90% in various research papers.
- Research often focused on limited departments like IT or finance.
- Behavioral, technical, and productivity-related factors were considered.
- Challenges Observed
    - Data inconsistency
    - Lack of real-time prediction
    - Incomplete feature sets
- Grouped models worked better than single models.

Algorithm Used

- **Decision Tree:** Simple and easy to understand.
- **KNN:** Finds similar data points but can be slow.
- **Naive Bayes:** Good with yes/no type data.
- **Neural Networks:** Works well but needs a lot of data.

Advantages & Disadvantages

- **Advantages**
- Fast and automatic process
- Gives correct and fair results
- Works for many employees at once
- Removes personal bias
- **Disadvantages**
- Needs a lot of good data
- Can carry bias if data is not fair
- Needs skilled people to manage
- Raises privacy concerns

Uses Of Machine Learning

- HR can use it for yearly reviews
- Helps in deciding who needs training
- Finds employees who are struggling
- Helps in promotions and rewards
- Useful for planning teams and roles
- Can be shown on HR dashboards

Conclusion

- ML helps in fair and fast performance checks
- Reduces workload for HR
- Helps in planning for the future
- Needs updates with new data often
- Must be used responsibly
- Can be improved with real-time systems

References

- IEEE and Springer articles on ML and HR
- Kaggle datasets about employee data
- Book: Hands-On ML by Aurélien Géron
- Google Scholar and ResearchGate papers
- Real company examples and case studies